AI-Powered Dyslexia Support System

Project Team

Abubakar Shahid 22i-1883
Muhammad Ali Zaib 22i-1900
Syed Abdur Rahman Grami 22i-2008

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Supervised by

Dr. Hassan Mujtaba

Co-Supervised by

Ms. Saira Qamar

Department of Data Science
National University of Computer and Emerging Sciences
Islamabad, Pakistan

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Abstract

This proposal presents DyslexAI, an AI-powered system designed to support learners with dyslexia by transforming their handwritten work into clear, corrected, and personalised outputs. The workflow begins with image preprocessing and a hybrid OCR pipeline (character- and word-level) to transcribe dyslexic handwriting into digital text. A correction module, based on sequence-to-sequence models and dyslexia-specific error patterns, refines the text to improve readability while preserving context.

Beyond correction, the system introduces several assistive features: error highlighting to raise awareness of mistakes, an adaptive personalisation engine that builds learner-specific error profiles, and gamified exercises to encourage consistent practice through rewards and progressive challenges. A feedback system provides targeted suggestions, while a dashboard with dyslexia-friendly fonts visualises accuracy, trends, and improvement over time. An experimental dyslexia detection module, exploring error-pattern and eye-tracking analysis, is also included as an exploratory feature (non-clinical).

The system will be deployed as lightweight web and mobile applications, ensuring accessibility for learners, parents, and educators. Development is structured into four phases (Sep–Oct 2025, Nov–Dec 2025, Feb–Mar 2026, Apr–May 2026), each aligned with specific milestones and evaluation metrics. By combining automated recognition, correction, personalisation, and engagement-driven practice, DyslexAI aims to provide an innovative, accessible tool that strengthens writing skills and confidence in dyslexic learners.

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Chapter 1

Introduction

1.1 Problem Statement

Dyslexia is a neurodevelopmental disorder that impairs reading, writing, and spelling, often visible in handwriting as letter reversals (e.g., *b/d*, *p/q*), spacing issues, and phonetic-based spelling errors. It affects nearly 15–20% of the population [1], creating persistent challenges in learning and self-expression.

Current tools like standard OCR systems and spell-checkers fail to address dyslexic handwriting because they are trained on typical writing patterns. They miss dyslexia-specific errors, provide inaccurate corrections, and lack personalization. This results in delayed feedback, reduced learner confidence, and extra burden on teachers who must manually review and correct student work.

In short, existing OCR and spell-check tools fail to accurately recognize and support dyslexic handwriting, leaving learners without personalized, effective assistance.

1.2 Motivation

Dyslexia affects an estimated 15– 20% of the population, yet existing support systems are inefficient and time-consuming. According to the British Dyslexia Association, early and targeted feedback can improve handwriting skills by up to 30% [2]. However, teachers often spend 5–10 hours per week per class manually reviewing student work. This creates a considerable burden while still leaving room for inconsistencies.

An AI-driven tool that can process handwritten images, correct text, highlight errors, and provide personalised feedback offers a scalable solution to this challenge. By automating repetitive tasks, such a system can save valuable instructional time, deliver consistent evaluations, and enhance learning outcomes.

To date, generic handwriting recognition tools have been developed and trained primarily on typical handwriting datasets (e.g., MNIST, IAM), which do not capture the distortions characteristic of dyslexic writing. Meanwhile, studies in developmental psychology highlight handwriting features such as letter reversals, spacing inconsistencies, and phonetic spelling errors as common in dyslexia [3]. This suggests that existing OCR systems are unlikely to perform effectively for dyslexic learners — a gap that our work addresses.

By integrating dyslexia-specific datasets, adaptive learning, and gamified exercises, our system directly addresses this gap. A specialised solution ensures higher accuracy, empowers learners with personalised practice, and fosters independence and confidence in students struggling with dyslexia.

1.3 Problem Solution

To address the identified challenges, we propose an end-to-end pipeline designed to handle input variability and deliver corrective, personalised feedback effectively. The solution consists of the following stages:

1. **Image Upload:** A photo or scan of the handwritten page is provided as input.
2. **Preprocessing and Segmentation:** Using OpenCV, noise is reduced and the image is segmented into lines, words, and characters.
3. **Handwriting Recognition:** A hybrid OCR approach (character-level + word-level) fine-tuned on dyslexic handwriting datasets, to capture distortions and spacing issues.
4. **Text Assembly:** Recognized words are combined to reconstruct sentences and paragraphs, preserving the original structure of the writing.
5. **Text Correction:** A fine-tuned seq2seq model (e.g., T5-small) [4] corrects text by modelling common dyslexic error patterns such as phonetic substitutions and reversals, thereby improving readability.
6. **Error Highlighting:** The corrected text is paired with highlighted mistakes for learner awareness.
7. **Feedback Generation:** Rules and learned patterns are applied to generate targeted recommendations.
8. **Adaptive Personalisation:** Learner error profiles are built to provide a customised practice pathway.
9. **Gamified Exercises:** Tracing, rewriting, and correction tasks with rewards to encourage engagement.

10. **Progress Visualisation:** A dashboard provides metrics such as error rates, trends, and frequent mistakes.

11. **Dyslexia Detection (Exploratory):** An optional module investigates detection using handwriting error patterns or eye-tracking, for non-medical educational insights.

The overall workflow of the system is illustrated below.

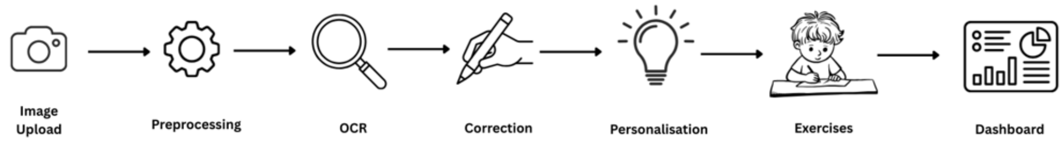

Figure 1.1: Proposed solution workflow for DyslexAI.

1.4 Objectives

- Attain high character recognition accuracy on dyslexic handwriting samples, validated against established OCR benchmarks.
- Enhance text readability through targeted fine-tuning, measured using CER and WER improvements over raw OCR outputs.
- Provide explainable and personalised feedback to help reduce recurring errors, evaluated using learner performance metrics across sessions.
- Support adaptive learning through a personalisation engine and gamified practice to maintain learner motivation.
- Ensure efficient, low-latency processing suitable for practical classroom or home use.
- Explore experimental dyslexia detection methods for non-clinical, educational support.

1.5 Stakeholders

- **Primary:** Dyslexic learners (ages 8–18), who benefit from immediate, personalised feedback and engaging practice activities.
- **Secondary:** Educators and parents, who gain actionable insights and time savings for more targeted interventions.
- **Institutional:** Schools and dyslexia support centers, which can integrate the tool to provide scalable assistance, supported by evidence that technology-based interventions reduce dropout rates by 15–20%.

Chapter 2

Project Description

2.1 Scope

- **Handwriting correction** from uploaded images (English only).
- **OCR** with transformer-based and hybrid models (character-level + word-level fusion).
- **Text correction** using sequence-to-sequence models (with T5 as a primary candidate, while considering alternatives).
- **Adaptive personalisation engine** for building user error profiles and recommending targeted exercises.
- **Gamified practice exercises** (letter tracing, rewriting, correction tasks) with points, badges, and levels.
- **Error highlighting** for marking common handwriting mistakes (e.g., reversals, spacing, spelling).
- **Feedback system** with constructive suggestions for improvement.
- **Analytics dashboard** with progress tracking, error rates, and personalized insights.
- **Dyslexia-friendly fonts** and accessibility features in the interface.
- **Experimental dyslexia detection module** (non-medical, using error patterns or eye-tracking).
- **Functional web and mobile applications** for accessibility.

2.1.1 Deliverables

- Working web and mobile applications.
- Fine-tuned OCR and correction models, with documentation of model comparison and selection.
- Adaptive personalisation engine with dynamic learning paths.
- Gamified practice module with progressive difficulty and rewards.
- Error highlighting and integrated feedback system.
- Dyslexia-friendly fonts and accessible UI design.
- Analytics dashboard with detailed reporting and progress visualization.
- Experimental dyslexia detection prototype (exploratory).
- Evaluation on test data with CER/WER and readability metrics.
- User guide and technical documentation.

2.2 Modules

1. **Handwriting Recognition Module**
    - Recognizes handwritten input at both character level (normal, reversed, corrected letters) and word level (words, sentences).
    - Uses a hybrid approach combining character- and word-level recognition through fusion logic trained on dyslexia handwriting datasets [5].
    - Helps detect common dyslexic handwriting issues such as letter reversals, misformed characters, and irregular writing styles.
2. **Error Detection & Highlighting Module**
    - Analyzes recognized handwriting for dyslexia-specific patterns (reversals, spacing issues, phonetic spellings).
    - Highlights errors in the uploaded handwriting so learners can easily identify their mistakes.
    - Assists teachers and parents in understanding exactly where the child struggles.

3. **Error Correction Module**

   - Suggests corrected versions of text while preserving the learner's intended meaning.
   - Helps students improve their writing by guiding them toward more accurate outputs.
   - Similar approaches are found in tools like Dyslexico [6] and LaMPost [7]

4. **Adaptive Personalisation Module**

   - Builds a personal error profile by monitoring user mistakes over time.
   - Recommends targeted exercises that focus on frequent problem areas (e.g., letter reversals, spacing).
   - Continuously updates difficulty and recommendations as the learner improves.
   - Creates a custom learning path instead of fixed practice sets.

5. **Exercise & Gamification Module**

   - Provides practice tasks such as letter/word tracing, rewriting, and error correction activities.
   - Builds on the idea of assistive educational apps for dyslexia [8].
   - Exercises are adaptive in difficulty, adjusting automatically to student performance.
   - Includes gamification features such as points, badges, and levels to keep learners motivated and engaged.

6. **Feedback System Module**

   - Offers instant feedback after handwriting recognition and practice exercises.
   - Clearly shows what was correct, what was wrong, and how to improve.
   - Builds learner confidence by giving timely and consistent guidance.

7. **Dashboard Module**

   - Tracks student progress across different sessions.
   - Provides visual insights such as charts and progress bars on accuracy, common mistakes, and improvements.
   - Allows teachers and parents to monitor multiple learners simultaneously.
   - Functions as the central reporting hub of the application.

8. **User Interface (Web/Mobile App) Module**
    - Offers a user-friendly platform for uploading handwriting, viewing corrections, and practicing exercises.
    - Designed with dyslexia-friendly features such as readable fonts, accessible layouts, and suitable color schemes.
    - Ensures smooth integration of all system features for students, teachers, and parents.
9. **(Optional/Future) Eye-Tracking & Reading Analysis Module**
    - Uses webcam-based gaze tracking to study reading behavior such as fixations, regressions, and irregular eye movements.
    - Identifies reading difficulties in real time, providing an additional method for dyslexia detection.
    - Complements handwriting analysis for a more holistic learning support system.

2.3 Tools & Technologies (Tentative)

- **Programming & AI:** Python, TensorFlow / PyTorch, Scikit-learn, OpenCV
- **Web & App Development:** React.js, Node.js / Django
- **Database & Storage:** SQL, Firebase, Pandas
- **Visualization:** Chart.js, Tailwind CSS
- **Deployment:** Docker, AWS / Heroku / Firebase

2.4 Work Distribution

Member	Responsibilities
Ali Zaib	Preprocessing pipeline, Handwriting Recognition Module, Handwriting Correction Module, Personalisation Engine (adaptive learning), Dashboard, and System Integration across modules.
Abubakar Shahid	Data collection and integration, Handwriting Recognition Module, Handwriting Correction Module, Feedback System and Error Highlighting, Web Application, and System Integration across modules.
Abdur Rahman Grami	Data organisation and augmentation, Handwriting Recognition Module, Handwriting Correction Module, Gamified Exercises, Dyslexia Detection, and Mobile Application.

2.5 Timeline

Sep–Oct 2025	Data preprocessing and cleaning; dataset augmentation; hybrid OCR (character + word-level recognition).
Nov–Dec 2025	Handwriting correction module; adaptive personalisation engine.
Feb–Mar 2026	Gamified practice exercises; error highlighting; feedback system; analytics dashboard.
Apr-May 2026	Experimental dyslexia detection prototype; complete system integration (web + mobile apps); testing and evaluation.

2.6 Future Work

- Expand to multilingual handwriting recognition and correction (e.g., Urdu).
- Enable real-time handwriting input from tablets/stylus devices.
- Improve dyslexia detection with advanced behavioral/eye-tracking studies.
- Large-scale validation with dyslexia support centers and schools.
- Integration with learning management systems (LMS) for broader adoption.

2.7 Conclusion

In conclusion, DyslexAI is an AI-powered handwriting support system tailored for dyslexic learners. It combines a hybrid OCR pipeline with correction models to handle dyslexia-specific challenges such as reversals and spacing errors. Beyond correction, the system offers error highlighting, adaptive personalisation, gamified practice, and a dashboard with dyslexia-friendly fonts. An exploratory dyslexia detection module is also included as a non-clinical feature. By integrating recognition, correction, personalisation, and progress monitoring, DyslexAI provides a scalable, accessible tool with both academic value and meaningful social impact.

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